ECA5601 –Advanced Wireless Communication Systems / 5G / 6G
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Experiment: 6

5G NR Frame Structure Analysis with Different Subcarrier Spacing DATE:12-8-26

Aim

To analyze the 5G NR frame structure by studying the effect of different subcarrier spacings, slot durations, and OFDM symbols using MATLAB.

Objectives:

To analyze different Subcarrier Spacing (SCS) values used in 5G NR numerology.

Software Required: • MATLAB

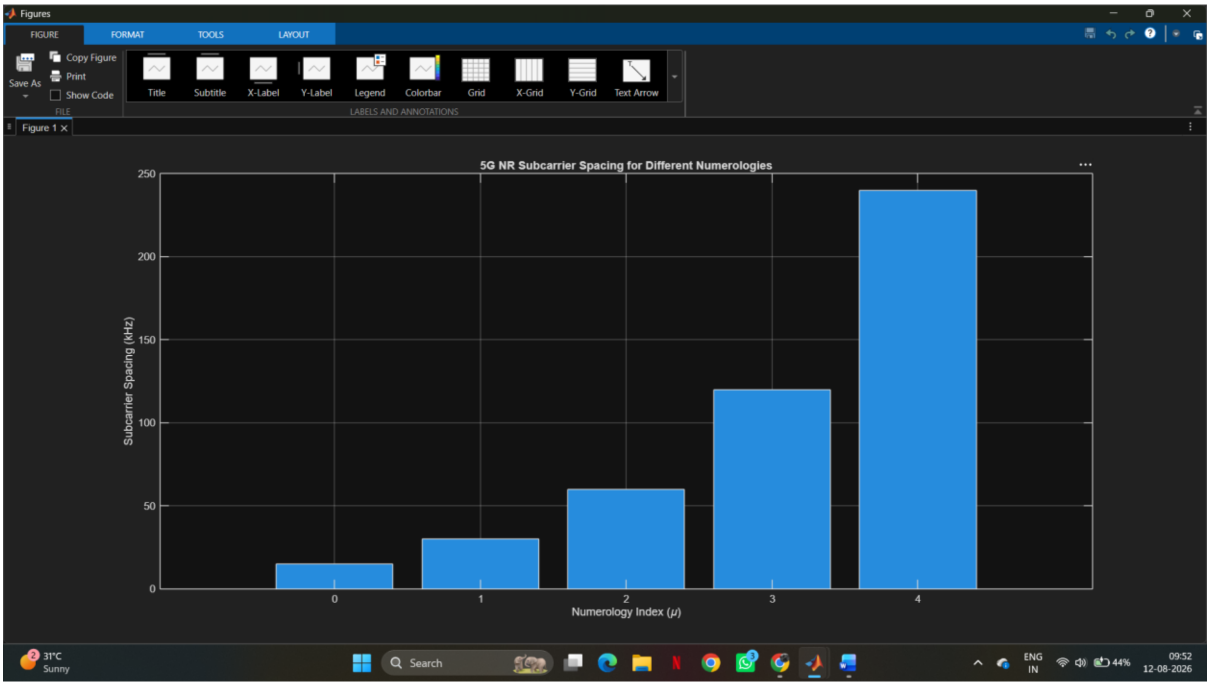
Procedure:

- Define the supported 5G NR subcarrier spacing values.
- Display the values using MATLAB.
- Plot the subcarrier spacing for different numerologies.

MATLAB Program:

```
clc;
clear;
close all;
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
mu = 0:4; % Numerology Index
bar(scs)
grid on
set(gca,'XTick',1:5)
set(gca,'XTickLabel',mu)
xlabel('Numerology Index (\mu)')
ylabel('Subcarrier Spacing (kHz)')
title('5G NR Subcarrier Spacing for Different Numerologies')
```

OUTPUT:



Result:

The MATLAB analysis successfully demonstrated the flexible numerology of 5G NR. The results showed that increasing the subcarrier spacing decreases the slot duration while maintaining a constant number of OFDM symbols per slot under the normal cyclic prefix configuration.